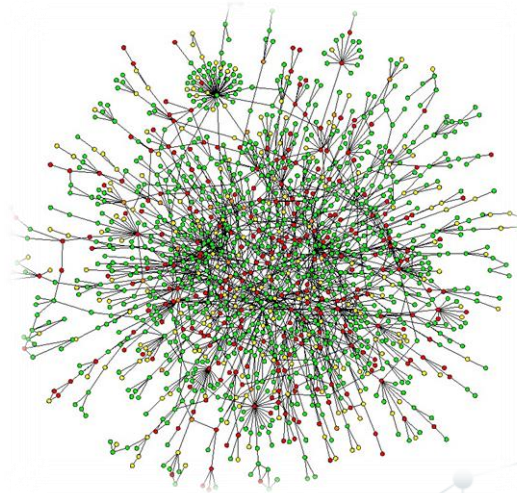
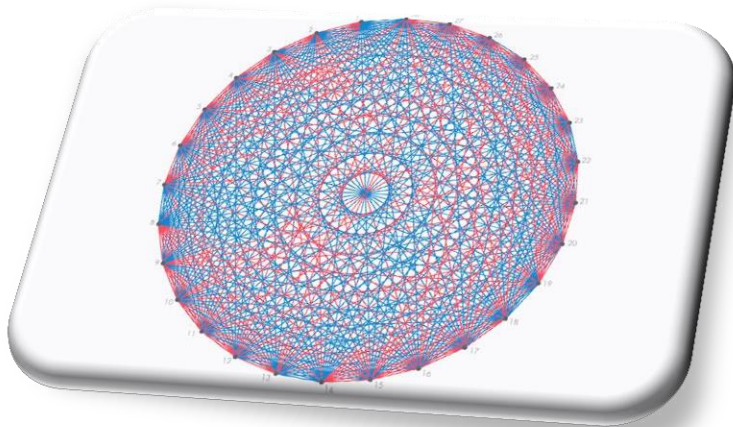
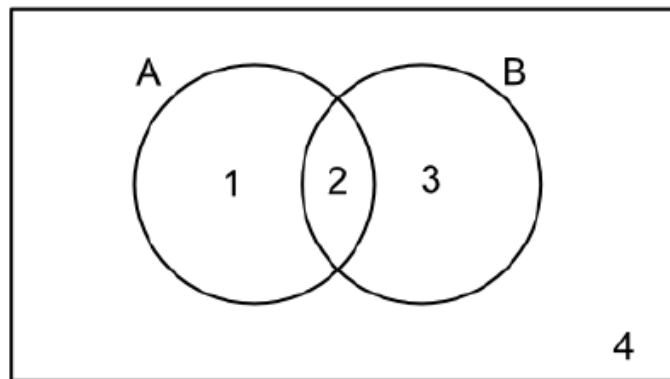


GRAPHICAL MODELS



RECAP



$$P(A^c) = 1 - P(A) \quad (1)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad (2)$$

$$P(A \cup B) \leq P(A) + P(B) \quad (3)$$

RECAP

The expectation (mean) of a random variable X can be expressed as

$$E(X) = X_{\text{mean}} = \sum_{x \in \mathcal{X}} xP(X = x). \quad (4)$$

The variance and covariance can be defined therefore in terms of the expectation, where

$$\text{Var}(X) = E((X - E(X))^2) = E(X^2) - E(X)^2, \quad (5)$$

$$\text{Cov}(X) = E(XY) - E(X)E(Y). \quad (6)$$

Conditional expectations and variances follow from the conditional distributions

$$E(X | Y = y) = \sum_{x \in \mathcal{X}} xP(X = x | Y = y). \quad (7)$$

RECAP

Two random variables are independent when the probability distribution of one random variable does not affect the other. More concretely, two random variables X and Y are independent, that is, $X \perp Y$, if and only if

$$P(X = x, Y = y) = P(X = x)P(Y = y), \quad \text{for all } x \in \mathcal{X}, y \in \mathcal{Y}. \quad (8)$$

If X and Y are continuous with joint density function $f_{X,Y}(x, y)$, then the above condition reduces to finding functions $h(x)$ and $g(y)$ such that

$$f_{X,Y}(x, y) = h(x)g(y). \quad (9)$$

RECAP

Two random variables X and Y are conditionally independent given a third variable Z , denoted as $X \perp Y \mid Z$, if and only if

$$P(X = x, Y = y \mid Z = z) = P(X = x \mid Z = z)P(Y = y \mid Z = z), \quad (10)$$

for all $x \in \mathcal{X}, y \in \mathcal{Y}, z \in \mathcal{Z}$.

This is equivalent to saying

$$P(X = x \mid Y = y, Z = z) = P(X = x \mid Z = z).$$

Note that $X \perp Y \mid Z$ does not imply that $X \perp Y$, and vice versa.

GRAPH THEORY PRELIMINARIES

A graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ consists of:

- A set of nodes/vertices $\mathcal{V} = \{1, 2, \dots, n\}$. Illustrated by a dot or cross.
- A set of edges $\mathcal{E} = \{(u, v) \mid u, v \in \mathcal{V}\}$, which consists of node pairs. This is illustrated by a line connecting the two nodes in the node pair.
- If the pairs in $\mathcal{E}$ are unordered, that is $(u, v) = (v, u)$, then graph is called undirected (the lines in the graph have no directional arrows). If the pairs are ordered, that is $(u, v) \neq (v, u)$, then the graph is called directed (the lines in the graph has arrows on them).
- We will be dealing with *simple* graphs here, which means we do not allow self loops and multiple edges between the same node pairs. We also will not be considering graphs with a mixture of ordered and unordered edges.

DIRECTED AND UNDIRECTED GRAPHS

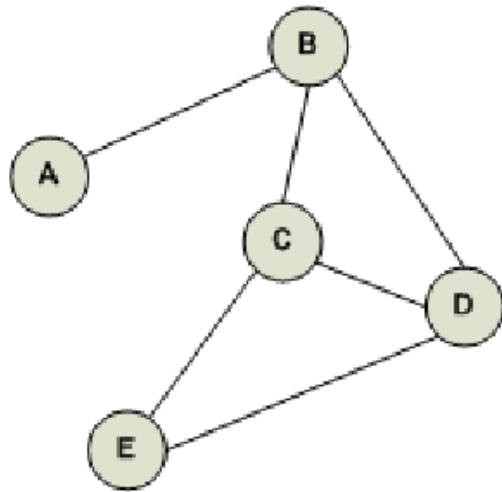


Fig 1. Undirected Graph

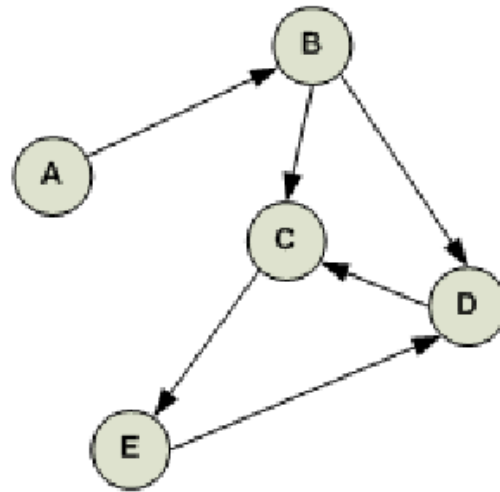


Fig 2. Directed Graph

SEPARATED NODE SETS

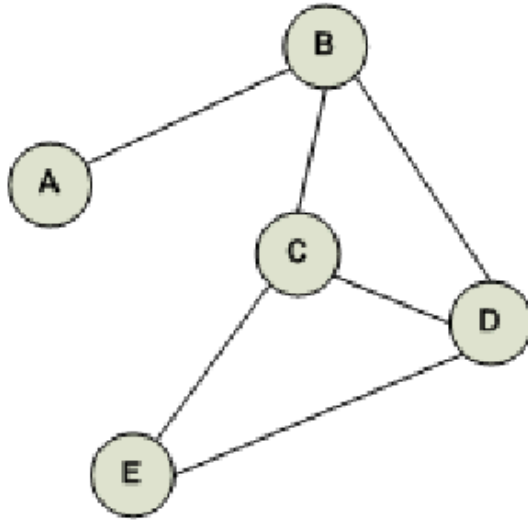


Fig 1. Undirected Graph

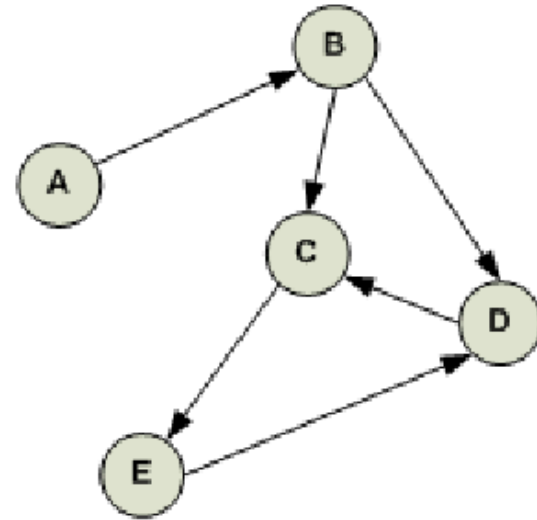
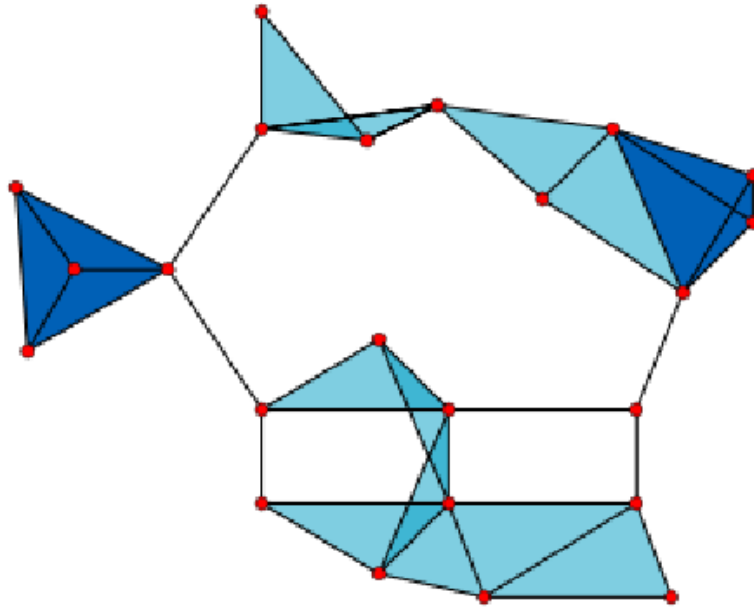


Fig 2. Directed Graph

CLIQUE



GRAPHICAL MODELS

A graphical model is a multivariate probability distribution associated with a graph representation, where the nodes of the graph represent the variables in the distribution and the edges represent the relationships between nodes.

If the associated graph is directed (undirected), then the graphical model is directed (undirected).

In this lecture, we will be discussing undirected graphical models.

UNDIRECTED GRAPHICAL MODELS

Which is more important, independence or conditional independence?

One main type of undirected graphical model is known as the Markov random field (MRF).

In a MRF, if no edge exists between two nodes u and v , then X_u must be conditionally independent of X_v given the rest of the variables.

MARKOV RANDOM FIELDS

Markov Properties on Undirected Graphs

Fact: (F) $\Rightarrow$ (G) $\Rightarrow$ (L) $\Rightarrow$ (P).

(F) P factorizes according to G : $p(x) = \prod_{C \in \mathcal{C}} \phi_C(x_C)$.

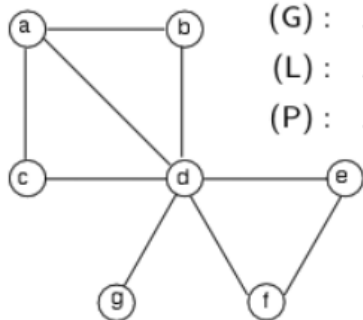
(G) For any disjoint subsets A, B, S of V , $A \perp B | S \Rightarrow X_A \perp X_B | X_S$.

(L) For all $v \in V$, $X_v \perp X_{V \setminus \text{cl}(v)} | X_{\text{bd}(v)}$.

(P) For all pairs of non-adjacent vertices (v, v') in G , $X_v \perp X_{v'} | V \setminus \{v, v'\}$.

Illustration:

G :



(F) : $p(x) = \phi_1(x_a, x_b, x_d) \phi_2(x_a, x_c, x_d) \phi_3(x_d, x_e, x_f) \phi_4(x_d, x_g)$

(This is the most general form of p that satisfies (F).)

(G) : $X_{\{a,b\}} \perp X_{\{e,g\}} | X_d$, $X_{\{c,g\}} \perp X_{\{b,f\}} | X_{\{a,d\}}$, etc.

(L) : $X_a \perp X_{\{e,f,g\}} | X_{\{b,c,d\}}$, $X_g \perp X_{\{a,b,c,e,f\}} | X_d$, etc.

(P) : $X_a \perp X_e | X_{\{b,c,d,e,f\}}$, $X_c \perp X_b | X_{\{a,d,e,f,g\}}$, etc.

MULTIVARIATE GAUSSIAN

The Gaussian distribution $\mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ has the density function,

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^p |\boldsymbol{\Sigma}|}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Omega} (\mathbf{x} - \boldsymbol{\mu}) \right\}, \quad (11)$$

where $\boldsymbol{\Omega} = \boldsymbol{\Sigma}^{-1}$, The edges of the associated graph $\mathcal{G}$ are exactly those corresponding to the nonzero entries of $\boldsymbol{\Omega}$, the precision matrix.

EXAMPLE

Let $\mathbf{X} = (X_1, X_2, X_3)$ be a multivariate Gaussian distribution with zero mean and covariance matrix

$$\mathbf{\Sigma} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & -1 \\ 0 & -1 & 3 \end{bmatrix}, \quad \mathbf{\Omega} = \begin{bmatrix} 2 & -3 & -1 \\ -3 & 6 & 2 \\ -1 & 2 & 1 \end{bmatrix}.$$

Observe that $X_1 \perp X_3$, or $X_1 \perp X_3 \mid \mathbf{X}_S$, where S is the null set. However, the graph according to the precision matrix is the complete graph, as the precision matrix has no off-diagonal zero entries. So while graph separability implies conditional independence, the converse does not necessarily hold.

ISING MODELS

The Ising model $\mathbf{X} = (X_1, \dots, X_n)$, used in the study of ferromagnetism, has state space $\mathcal{X} = \{-1, +1\}^n$ and density function

$$P(\mathbf{X} = \mathbf{x}) = \frac{1}{Z} \exp \left\{ \sum_{ij} J_{ij} x_i x_j + \sum_i h_i x_i \right\}, \quad (12)$$

where the associated graph $\mathcal{G} = \mathcal{V}, \mathcal{E}$ has an edge between nodes i and j if and only if $J_{ij} \neq 0$.

The normalizing constant has the form

$$Z = \sum_{\mathbf{x} \in \mathcal{X}} \exp \left\{ \sum_{ij} J_{ij} x_i x_j + \sum_i h_i x_i \right\}. \quad (13)$$

GRAPH LEARNING

The graph learning problem involves learning the underlying graph structure given samples from the distribution.

In the case of the Gaussian graphical model, we want to learn the non-zero structure of the precision matrix given the sample covariance matrix.

Difficult problem especially when dimension n is large.

GRAPHICAL LASSO

From samples $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)}$, the sample covariance is defined by

$$S = \frac{1}{N-1} \sum_{i=1}^N \left(\mathbf{x}^{(i)} - \sum_{j=1}^N \mathbf{x}^{(j)} / N \right) \left(\mathbf{x}^{(i)} - \sum_{j=1}^N \mathbf{x}^{(j)} / N \right)^T. \quad (14)$$

The graphical lasso uses maximum likelihood and sparsity in the graph to estimate the precision matrix,

$$\hat{\Omega} = \min_{\Omega} -\log |\Omega| + \text{tr}(S\Omega) + \lambda \|\Omega\|_1. \quad (15)$$

APPLICATIONS

1. Feature Correlations
2. Visual Representation
3. Small Data Learning
4. Wireless Communication Networks
5. Topological Learning
6. Social Network Analysis
7. Risk Assessment
8. Contagion Theory
9. Graph Signal Processing
10. Graph Neural Networks

DIRECTED GRAPHICAL MODELS

$\mathcal{G} = (V, E)$ **directed graph**. V **vertices**. E **edges** (ordered pairs of vertices)

X, Y **adjacent** if $X \rightarrow Y$ edge. Y **child** of X . X **parent** of Y .

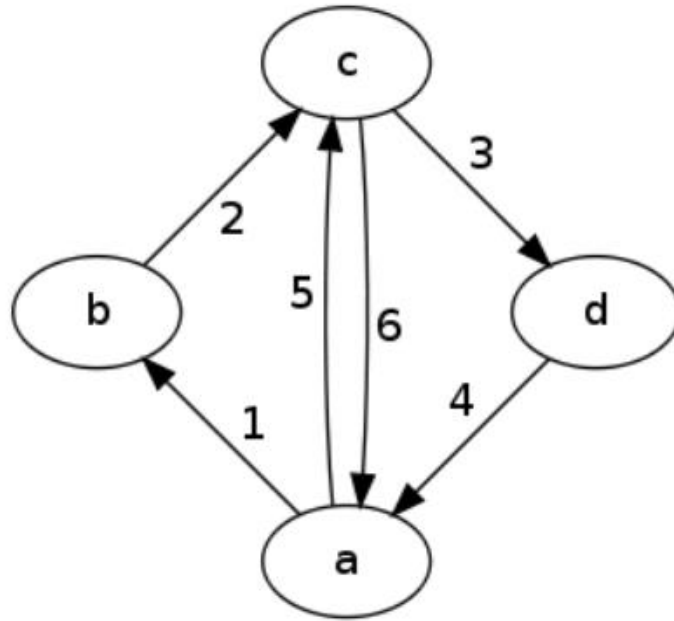
$X \rightarrow \dots \rightarrow Y$ **directed path**. Y **descendant** of X . X **ancestor** of Y .

$X \leftarrow \dots \rightarrow Y$ **undirected path** (ignore direction of arrows).

$X \rightarrow Y \leftarrow Z$ **collider**. $X \rightarrow Y \rightarrow Z$, $X \leftarrow Y \leftarrow Z$, $X \leftarrow Y \rightarrow Z$ **non-colliders**.

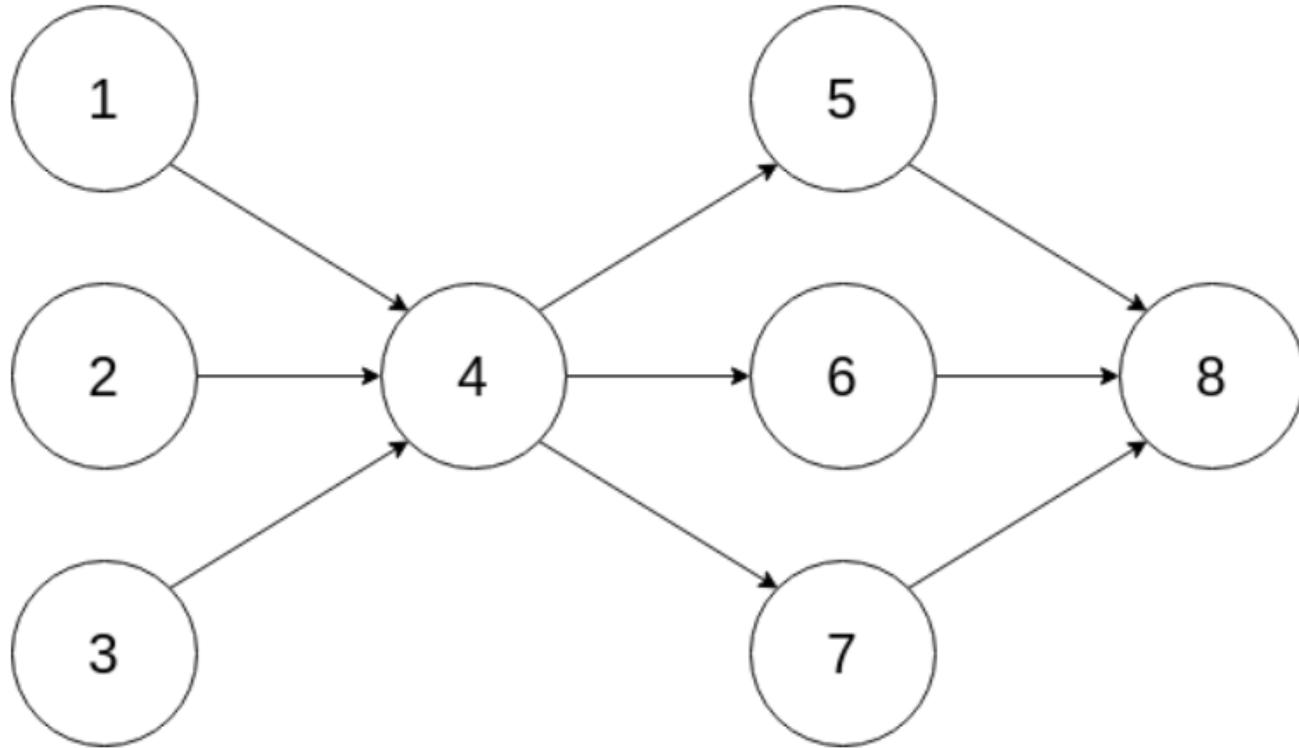
$X \rightarrow \dots \rightarrow X$ **cycle**. $\mathcal{G}$ **directed acyclic graph** if no cycles.

GRAPHS WITH CYCLES



$\{1, 2, 3, 4\}$ and $\{3, 4, 5\}$ are cycles.
 $\{3, 4, 6\}$ is not a cycle.

ACYCLIC GRAPHS



DIRECTED GRAPHICAL MODELS

A directed graphical model or Bayesian network consists of a multivariate random variable $\mathbf{X} = (X_1, \dots, X_n)$ and a corresponding graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where

- $\mathcal{V} = \{1, \dots, n\}$, where the variable X_i is represented by node i ,
- $(i, j) \in \mathcal{E}$ is denoted by an arrow connecting i to j ,
- the probability mass (density) function of $\mathbf{X}$ satisfies the factorization property.

FACTORIZATION

We denote the parents of node i by $\text{Pa}(i)$.

A probability mass function $P(\mathbf{X} = \mathbf{x})$ satisfies the factorization property with respect to a DAG if

$$P(\mathbf{X} = \mathbf{x}) = \prod_{i=1}^n P(X_i = x_i \mid \text{Pa}(i)). \quad (1)$$

EXAMPLE

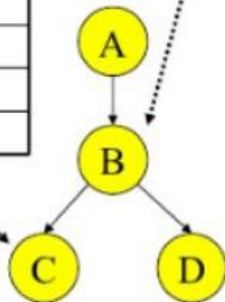
A Set of Tables for Each Node

A	P(A)
false	0.6
true	0.4

A	B	P(B A)
false	false	0.01
false	true	0.99
true	false	0.7
true	true	0.3

B	C	P(C B)
false	false	0.4
false	true	0.6
true	false	0.9
true	true	0.1

B	D	P(D B)
false	false	0.02
false	true	0.98
true	false	0.05
true	true	0.95



Each node X_i has a conditional probability distribution $P(X_i | \text{Parents}(X_i))$ that quantifies the effect of the parents on the node

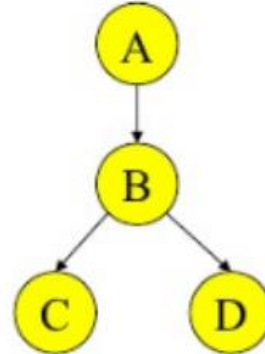
The parameters are the probabilities in these conditional probability tables (CPTs)

EXAMPLE

Using a Bayesian Network Example

Using the network in the example, suppose you want to calculate:

$$\begin{aligned} &P(A = \text{true}, B = \text{true}, C = \text{true}, D = \text{true}) \\ &= P(A = \text{true}) * P(B = \text{true} \mid A = \text{true}) * \\ &\quad P(C = \text{true} \mid B = \text{true}) P(D = \text{true} \mid B = \text{true}) \\ &= (0.4)*(0.3)*(0.1)*(0.95) \end{aligned}$$



EXAMPLE

EXAMPLES



Smoking

$$\begin{aligned} & f(\text{overweight}, \text{smoking}, \text{heart disease}, \text{cough}) \\ &= f(\text{overweight}) \times f(\text{smoking}) \\ &\times f(\text{heart disease} \mid \text{overweight}, \text{smoking}) \\ &\times f(\text{cough} \mid \text{smoking}). \end{aligned}$$

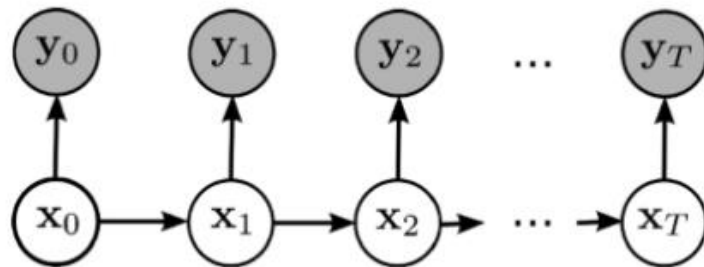
joint probability

root probabilities

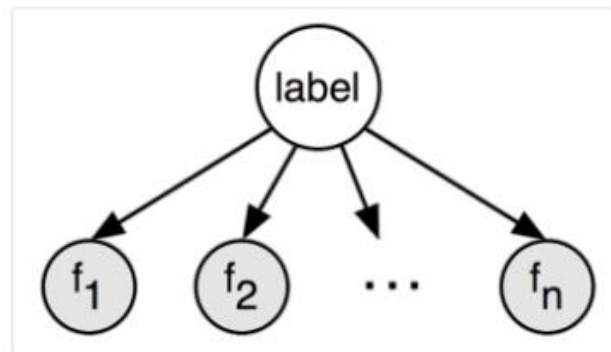
transition probabilities

EXAMPLE

Hidden Markov Model

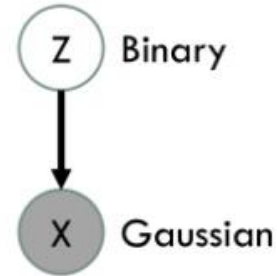


Naïve Bayes

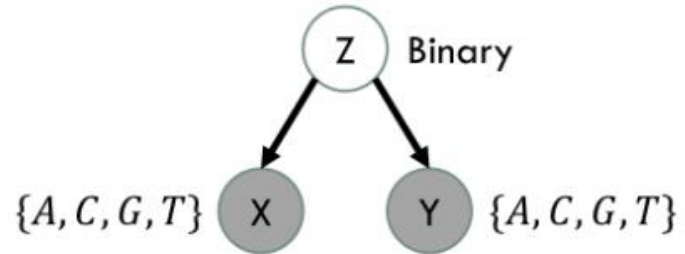


EXAMPLE

Gaussian Mixtures

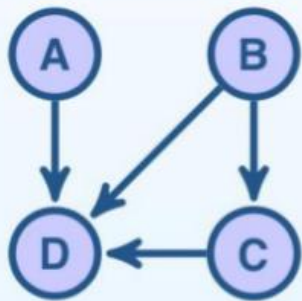


Phylogenetic Models



STRUCTURAL EQUATIONS

Parametrize the model using *structural equation modelling (SEM)*.



Gaussian

$$A = \varepsilon_A, \quad \varepsilon_A, \varepsilon_B, \varepsilon_C, \varepsilon_D \sim \mathcal{N}(0, 1)$$

$$B = \varepsilon_B$$

$$C = \lambda_{BC}B + \varepsilon_C$$

$$D = \lambda_{AD}A + \lambda_{BD}B + \lambda_{CD}C + \varepsilon_D$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -\lambda_{BC} & 1 & 0 \\ -\lambda_{AD} & -\lambda_{BD} & -\lambda_{CD} & 1 \end{pmatrix} \begin{pmatrix} A \\ B \\ C \\ D \end{pmatrix} \sim \mathcal{N}(0, \text{Id})$$

DISTRIBUTION

Let $\mathbf{K}$ be the matrix of the left hand side of the previous expression, and let $\mathbf{A} = (A, B, C, D)$ be the multivariate random variable. Then the graphical model in the previous slide implies that $\mathbf{K}\mathbf{A}$ is the standard normal distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$, which further implies that

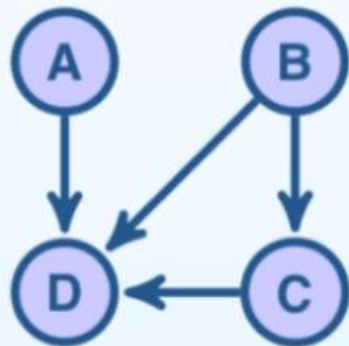
$$\mathbf{A} \sim \mathcal{N}\left(\mathbf{0}, \mathbf{K}^{-1} \left(\mathbf{K}^{-1}\right)^T\right). \quad (3)$$

LOCAL MARKOV PROPERTY

We say that a distribution $\mathbb{P}$ satisfies the **local Markov property** with respect to $\mathcal{G}$ if for all variables W ,

$$W \perp \tilde{W} \mid \pi_W$$

where π_W are the parents of W , and $\tilde{W}$ are the variables which are neither parents nor descendants of W .



$$A \perp B, C$$

$$B \perp A$$

$$C \perp A \mid B$$

$$D \perp \emptyset \mid A, B, C$$

D-SEPARATION

Consider the following undirected path from X to Z :

$$X - Y_1 - Y_2 - \dots - Y_n - Z.$$

Let W be some subset of vertices that do not contain X or Z .



Think of each intermediate vertex as a gate, and W a set of keys.

1. Collider gates are usually closed; all other gates are usually open.
2. If a collider or one of its descendants is in W , then that gate is opened.
3. If a non-collider is in W , then that gate is closed.

If all the gates are open, we say that X and Z are **d-connected** given W .

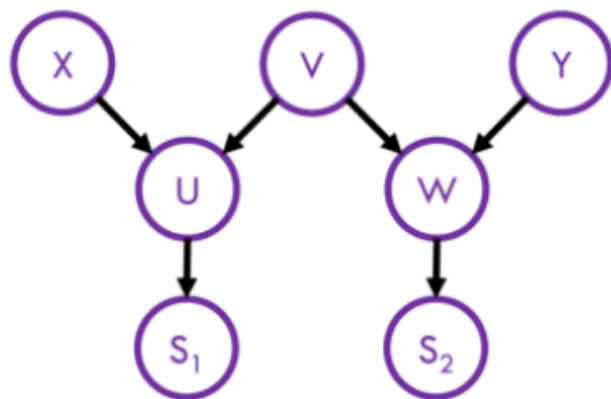
If we cannot find any such path, then X and Z are **d-separated** given W .

Sets S and T are **d-separated** given W if it is true for all $X \in S, Z \in T$.

EXAMPLE

EXAMPLES

Three layer DAG



X and Y are d-separated (given the empty set);

X and Y are d-connected given $\{S_1, S_2\}$;

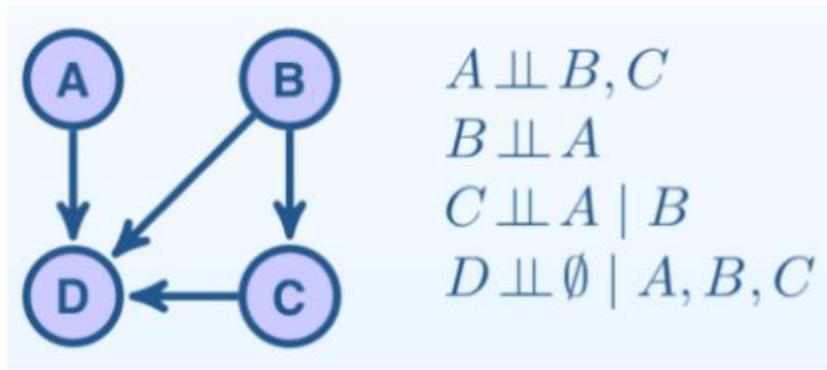
X and Y are d-separated given $\{S_1, S_2, V\}$.

GLOBAL MARKOV PROPERTY

We say that a distribution $\mathbb{P}$ satisfies the **global Markov property** with respect to $\mathcal{G}$ if

$$S \perp T \mid W$$

for all disjoint subsets S, T, W such that S and T are d-separated given W .



$$\begin{array}{l} A \perp B \mid C \\ A \perp C \end{array}$$

HAMMERSLEY-CLIFFORD THEOREM

The following are equivalent:

1. $\mathbb{P}$ satisfies the **factorization property** with respect to $\mathcal{G}$.
2. $\mathbb{P}$ satisfies the **local Markov property** with respect to $\mathcal{G}$.
3. $\mathbb{P}$ satisfies the **global Markov property** with respect to $\mathcal{G}$.

EXPLAINING AWAY

EXPLAINING AWAY



Why does conditioning on a collider lead to dependence?

If you don't know your friend is late:

$$\mathbb{P}(\text{Aliens}|\text{Watch}) = \mathbb{P}(\text{Aliens})$$

Aliens $\perp$ Watch

If you now know your friend is late:

$$\mathbb{P}(\text{Aliens}|\text{Watch}, \text{Late}) < \mathbb{P}(\text{Aliens}|\text{Late})$$

Aliens $\not\perp$ Watch | Late

Knowing his broken watch made him late **explains away** the possibility that he is late because he was abducted by aliens.

ADVANTAGES

- Conditional Independence relations can be read of the underlying DAG.
- Can describe causal relationships between variables.
- Can handle incomplete data or latent variables.